



THE UNIVERSITY OF
MELBOURNE

COMP90050 Advanced Database Systems

Semester 1, 2026

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Live lecture – Week 10





Today's topics

- Crash recovery summary
- ARIES example
- Q/A
 - Group presentations next week
 - Group report submission in Week 12
 - Final exam

We have seen crash recovery

What needs to be recovered if a crash happens?

- Transaction ended and has been made durable - good!
- If not durable – what additional information are needed to recover them?

Data pages in the buffer

Crash manager maintains both durability and atomicity

The changes by committed transactions – make them durable
The changes by aborted/running transactions - undo



What additional information we need to recover from a crash

Dirty Page table (WAL in place)

Pag e#	Oldest LSN (least Recent LSN)

X-table

Xid	Statu s	Last LSN

Log

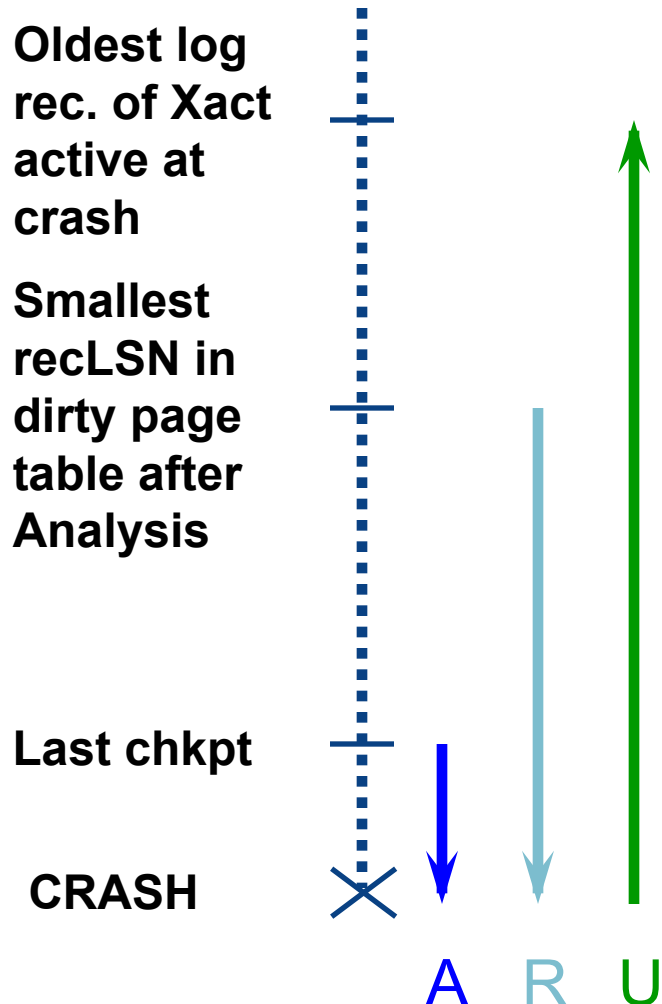
LSN	Tid	Type	P ag e	Le ng th	Of fs et	Old Value	New Value

Checkpoints: Periodically, the DBMS creates a checkpoint with current *Xact table* and *dirty page table*

Store logs and checkpoint records in a safe place



Crash Recovery: Big Picture

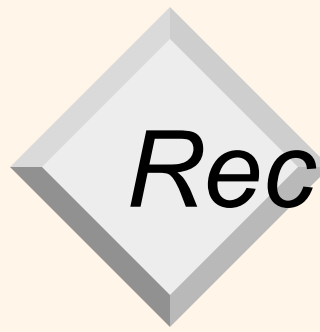


- ❖ Start from a **checkpoint** (found via **master** record).
- ❖ Three phases. Need to:
 - Figure out which Xacts committed since checkpoint, which failed (**Analysis**).
 - **REDO** *all* actions.
 - ◆ (repeat history)
 - **UNDO** effects of failed Xacts.



ARIES example: After a crash, we found these logs

```
0    BEGIN CHECKPOINT
5    END CHECKPOINT (EMPTY XACT TABLE AND DPT)
10   T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
15   T1: UPDATE P2 (OLD: WWW NEW: XXX)
20   T1: COMMIT, END transaction
```



Recovery: The Analysis Phase

X-table

Xid	Status	Last LSN
T1	R	
T2	R	

- ❖ Reconstruct state at checkpoint.
 - via **end_checkpoint** record.
- ❖ Scan log forward from checkpoint.
 - **End** record: Remove Xact from Xact table.
 - **Other records**: Add Xact to Xact table, set **lastLSN=LSN**, change Xact status on **commit**.
 - **Update** record: If P not in Dirty Page Table,
 - ◆ Add P to D.P.T., set its **recLSN=LSN**.

Dirty Page table

Page #	Oldest LSN (Recent LSN)
P5	
P6	
P7	



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

X-table

Xid	Status	Last LSN

Dirty Page table

Page #	Oldest LSN (Rec-LSN)

Analysis phase:

LSN 5: Initialize XACT table and DPT to empty.



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

X-table

Xid	Status	Last LSN
T1	R	10

Dirty Page table

Page #	Oldest LSN (Rec-LSN)
P1	10

Analysis phase:

LSN 5: Initialize XACT table and DPT to empty.

LSN 10: Add (T1, LSN 10) to XACT table. Add (P1, LSN 10) to DPT.



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

X-table

Xid	Status	Last LSN
T1	R	15

Dirty Page table

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Analysis phase:

LSN 5: Initialize XACT table and DPT to empty.

LSN 10: Add (T1, LSN 10) to XACT table. Add (P1, LSN 10) to DPT.

LSN 15: Set LastLSN=15 for T1 in XACT table. Add (P2, LSN 15) to DPT.



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

Dirty Page table

X-table

Xid	Status	Last LSN
T1	C	15

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Analysis phase:

LSN 5: Initialize XACT table and DPT to empty.

LSN 10: Add (T1, LSN 10) to XACT table. Add (P1, LSN 10) to DPT.

LSN 15: Set LastLSN=15 for T1 in XACT table. Add (P2, LSN 15) to DPT.

LSN 20: Change T1 status to "Commit" in XACT table. remove from



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

Dirty Page table

X-table

Xid	Status	Last LSN

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Analysis phase:

LSN 5: Initialize XACT table and DPT to empty.

LSN 10: Add (T1, LSN 10) to XACT table. Add (P1, LSN 10) to DPT.

LSN 15: Set LastLSN=15 for T1 in XACT table. Add (P2, LSN 15) to DPT.

LSN 20: Change T1 status to "Commit" in XACT table. remove from



Recovery: The REDO Phase

- ❖ We *repeat History* to reconstruct state at crash:
 - Reapply *all* updates (even of aborted Xacts!), redo CLR's.
- ❖ Scan forward from log rec containing smallest *recLSN* in D.P.T. For each CLR or update log rec *LSN*, REDO the action unless:
 - Affected page is not in the Dirty Page Table, or
 - Affected page is in D.P.T., but has *recLSN* > *LSN*, or
 - *pageLSN* (in DB) ≥ *LSN*.
- ❖ To REDO an action:
 - Reapply logged action.
 - Set *pageLSN* to *LSN*. No additional logging!



ARIES example

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- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

Dirty Page table

X-table

Xid	Status	Last LSN

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Redo phase:

Scan forward through the log starting at LSN 10.

LSN 10: Read page P1, check its PageLSN (in DB). If that $\text{PageLSN} < 10$, redo LSN 10 (set value to ZZZ) and set the page's $\text{PageLSN} = 10$.



ARIES example

- 0 BEGIN CHECKPOINT
- 5 END CHECKPOINT (EMPTY XACT TABLE AND DPT)
- 10 T1: UPDATE P1 (OLD: YYY NEW: ZZZ)
- 15 T1: UPDATE P2 (OLD: WWW NEW: XXX)
- 20 T1: COMMIT, END transaction

Dirty Page table

X-table

Xid	Status	Last LSN

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Redo phase:

Scan forward through the log starting at LSN 10.

LSN 10: Read page P1, check its PageLSN (in DB). If that PageLSN < 10, redo LSN 10 (set value to ZZZ) and set the page's PageLSN = 10.

LSN 15: Read page P2, check its PageLSN (in DB). If that



ARIES example

- 0 BEGIN CHECKPOINT
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- 20 T1: COMMIT, END transaction

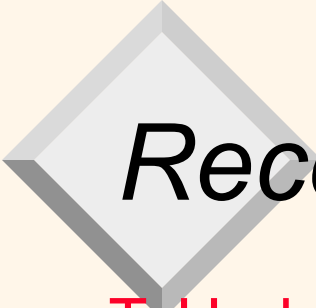
X-table

Xid	Status	Last LSN

Dirty Page table

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Undo phase:



Recovery: The UNDO Phase

ToUndo={ / | / is a lastLSN of a “loser” Xact }

We can form this list from Xtable.

Repeat:

- Choose the largest LSN among ToUndo.
- If this LSN is a CLR and undonextLSN==NULL
 - ◆ Write an End record for this Xact.
- If this LSN is a CLR, and undonextLSN != NULL
 - ◆ Add undonextLSN to ToUndo
 - ◆ (Q: what happens to other CLRs?)
- Else this LSN is an update. Undo the update, write a CLR, add prevLSN to ToUndo.

Until ToUndo is empty.



ARIES example

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- 20 T1: COMMIT, END transaction

X-table

Xid	Status	Last LSN

Dirty Page table

Page #	Oldest LSN (Rec-LSN)
P1	10
P2	15

Undo phase:

Do nothing; no transactions to undo.



Checkpoints

All DBMSs create checkpoints automatically – part of their job!

Some DBMSs provide functions to create checkpoints manually

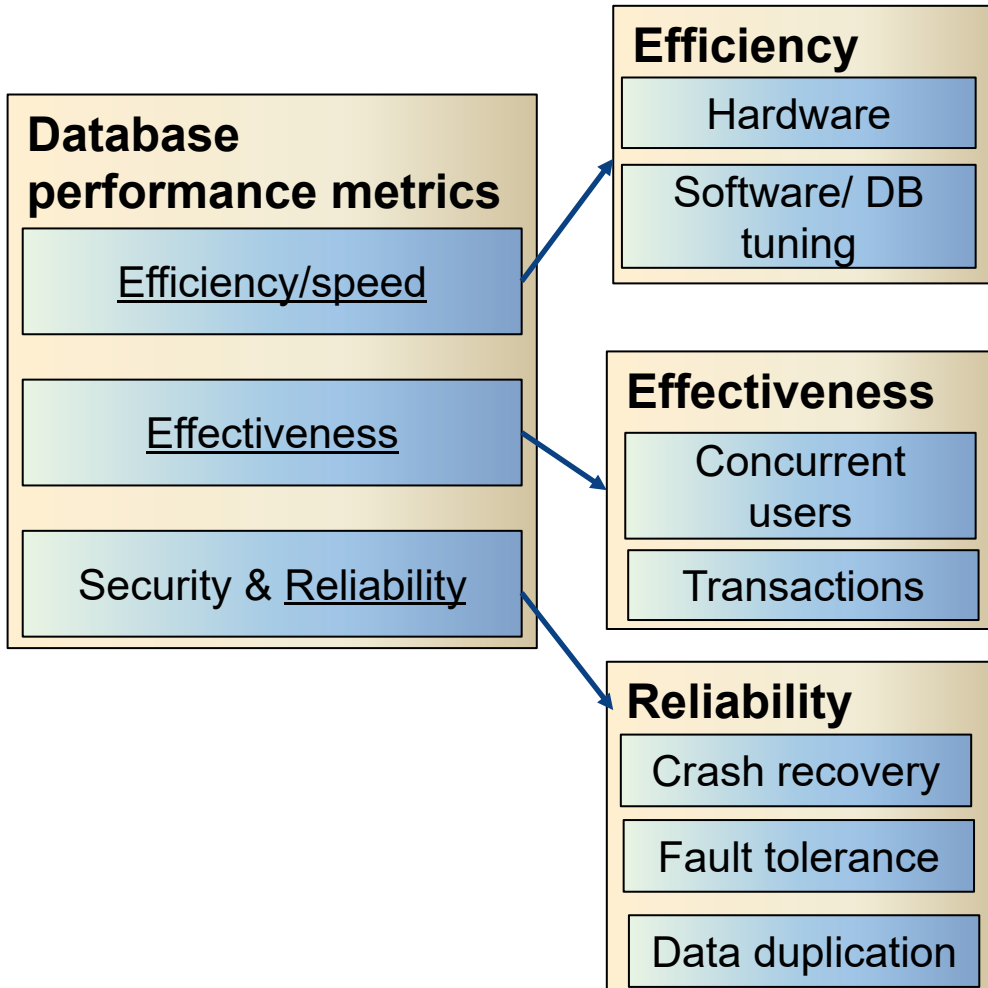
1. Indirect checkpoint - Issued in the background to meet the specified target recovery time for a database. Default is 1 minute.

```
ALTER DATABASE ... SET TARGET_RECOVERY_TIME = target_recovery_time {  
SECONDS | MINUTES }
```

2. Manual checkpoint – CHECKPOINT [*checkpoint_duration*]



Core Concepts of Database management system





Q/A

- Group presentations next week
- Group report submission in Week 12
- A sample final exam shared in canvas (the actual final exam will have more questions in it).